

6.2 Getting to Know the Basics

2. The `vxp_z` variable

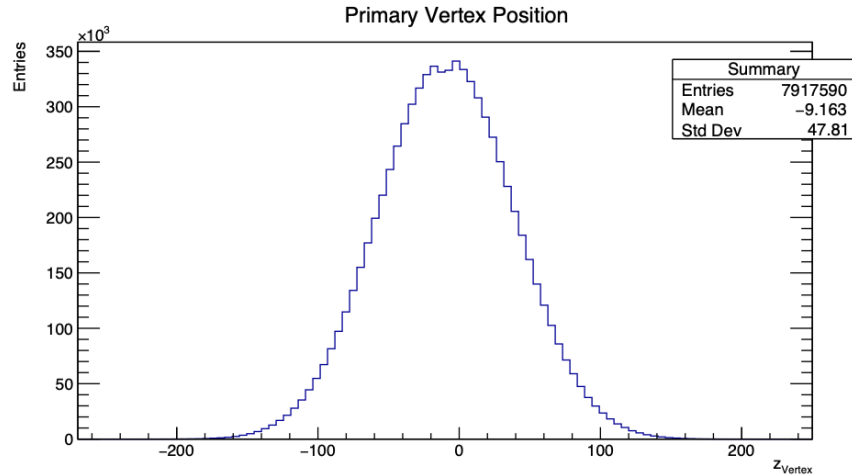


Figure 1: `vxp_z` graph

Plotting `vxp_z`: `vxp_z` is a float data type

it is the z-position of the primary vertex in mm

z-position is along the path of the particles, z axis along the beam pipe
center of the collider/collision is at $z = 0$

The graph shows number of measurements / entries at each position. It is a histogram with number of entries vs position of collision in millimeters.

We can read off the average position of the vertex (mean), as well as the standard deviation of the distribution.

Why this (mostly Gauss) distribution? Random variable is normally distributed.

Aim is to have collision at $z = 0$, different sources of random error combine to cause Gauss distribution of position.

Also Central Limit Theorem (CLT) means the sum of many random variables follows a distribution approaching a Gaussian.

3. Describe data

LHC Bunches are about 30 cm long, taking about 1 ns to travel their length.

They make up a beam of particles. It is easier to concentrate particles more closely when they are in bunches rather than a uniform beam.

They are spaced by 25 ns. For each beam 2808 bunches out of 3564 can contain protons.

Every filled bunch slot contains typically $N_i = O(10^{11})$ protons.

We see the distribution of the collision of two bunches.

Really there are two peaks very close to each other in the `vxp_z` graph. This could be because of the collision center moving between fills.

From website, we can see that for the 2012 run the collision center moved over time. Maybe this caused the two peaks.

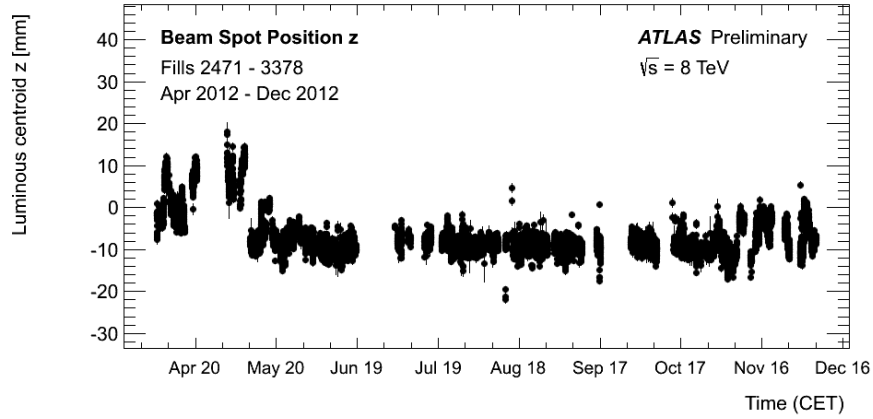


Figure 2: Beam Spot Position Graph from Website

Additionally, the distribution of protons inside a bunch is relevant. A plot of this would be helpful. If their distribution has two humps, so might the vertex position.

4. `lep_n` variable

Plotting `lep_n`: `lep_n` = Number of preselected leptons

The graph shows how many leptons were detected for the measurement/event. This is the number of counted leptons in the final state.

5. Describe data

In this part, we are talking only about how many leptons we actually *detected* in the final state. Therefore, neutrinos are ignored.

We expect 2 leptons in the final state. 1, 2 or 3 leptons in the final state are also possible, depending on the specific decay process.

In leading order we expect 2 leptons. The other numbers correspond to non-leading-order processes. Also, neutrinos are not detected, allowing odd numbers of leptons.

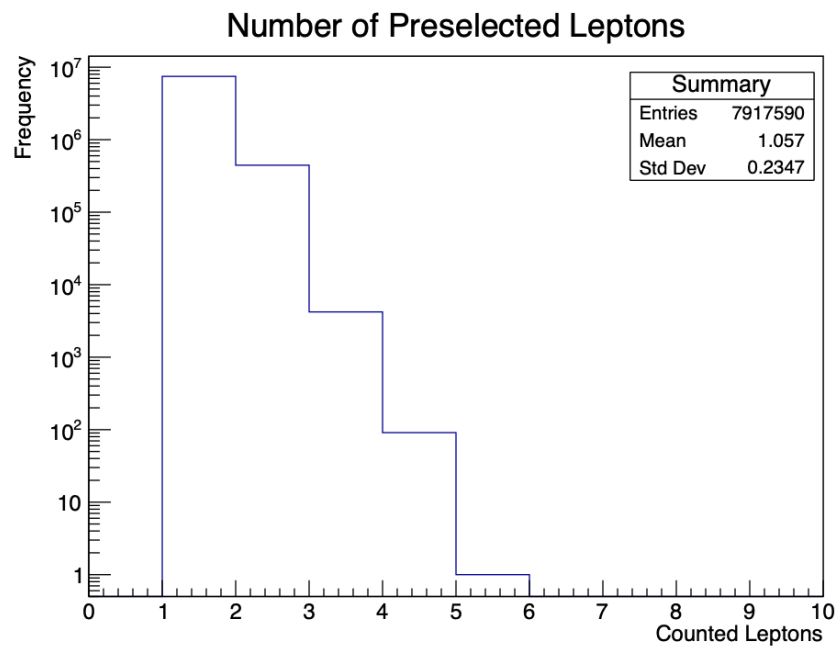


Figure 3: lep_n graph

Feynman diagrams of the 1, 2, 3 lepton processes go here!

The dominant process for 1-lep final:

$$\bar{u} + d \xrightarrow{W^-} e^- + \bar{\nu}_e$$

The dominant process for 2-lep final (q is any quark):

$$\bar{q} + q \xrightarrow{Z^0} e^- + e^+$$

The dominant process for 3-lep final (combine previous 2):

$$\bar{u} + d \longrightarrow e^- + \bar{\nu}_e + e^+ + e^-$$

6. lep_pt, lep_eta and lep_phi

Plotting lep_pt, lep_eta and lep_phi:

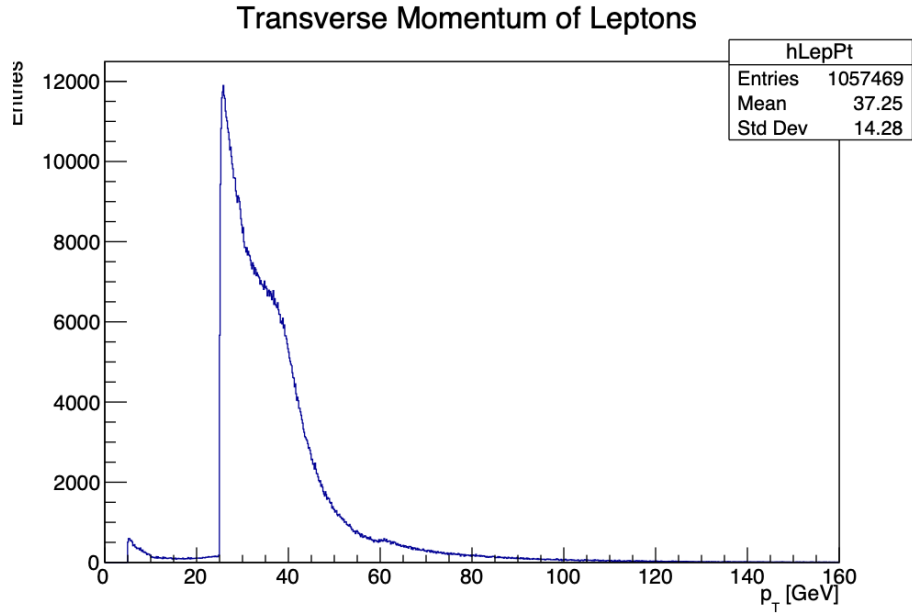
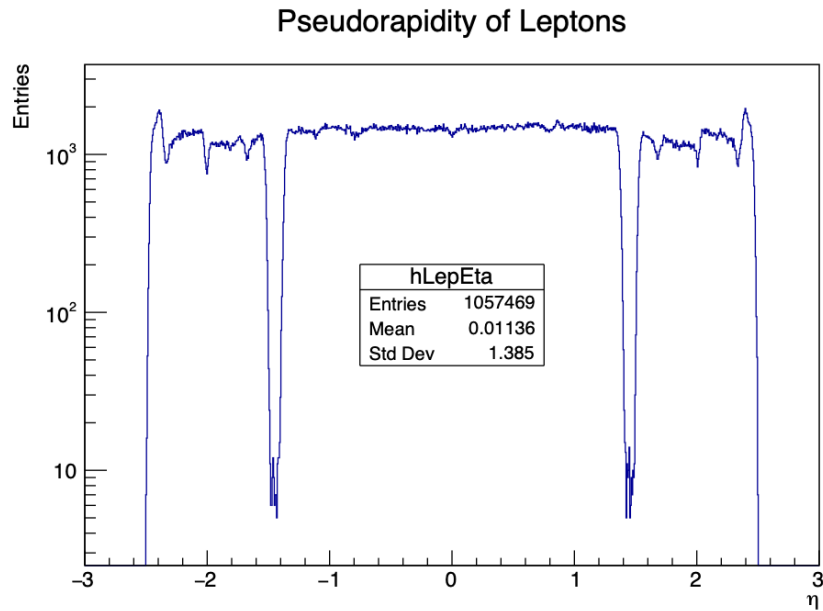
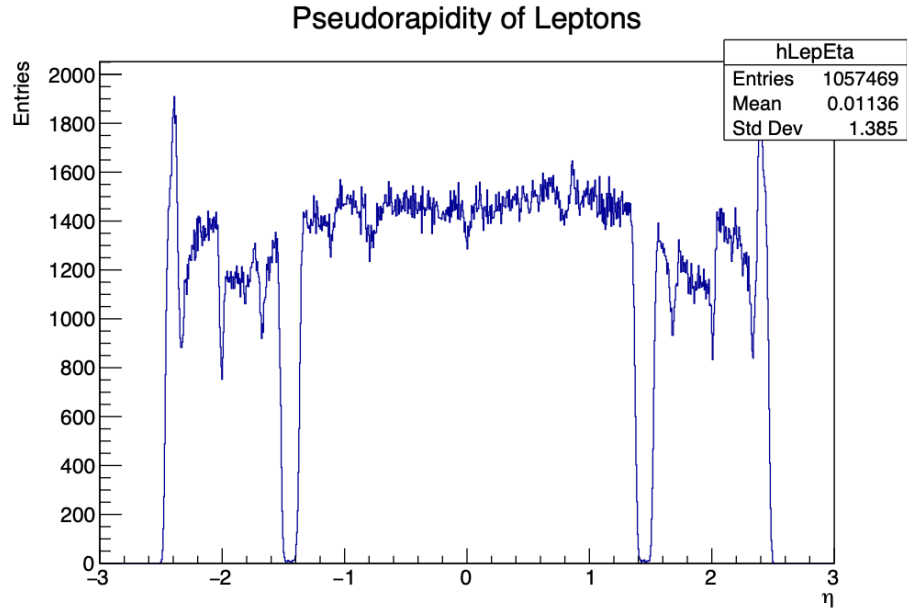


Figure 4: lep_pt graph

`lep_pt` is the Transverse momentum of the lepton (transverse to the beam direction)

The cutoff happens because an event must contain at least one lepton with $p_T > 25$ GeV. Otherwise, it is not recorded. This is a systematic cut off/*trigger* (not triggering means the event is not recorded/it is cut from the data).

The second cutoff is for events with two or more leptons, (at least one with $p_T > 25$ GeV). The other momenta measurements are not considered leptons if $p_T > 5$ GeV. An event like this is considered a 1 lepton event.



lep_eta is the Pseudorapidity of the lepton

Leptons are distributed uniformly in pseudorapidity. The gaps/cutoffs are because of detector geometry.

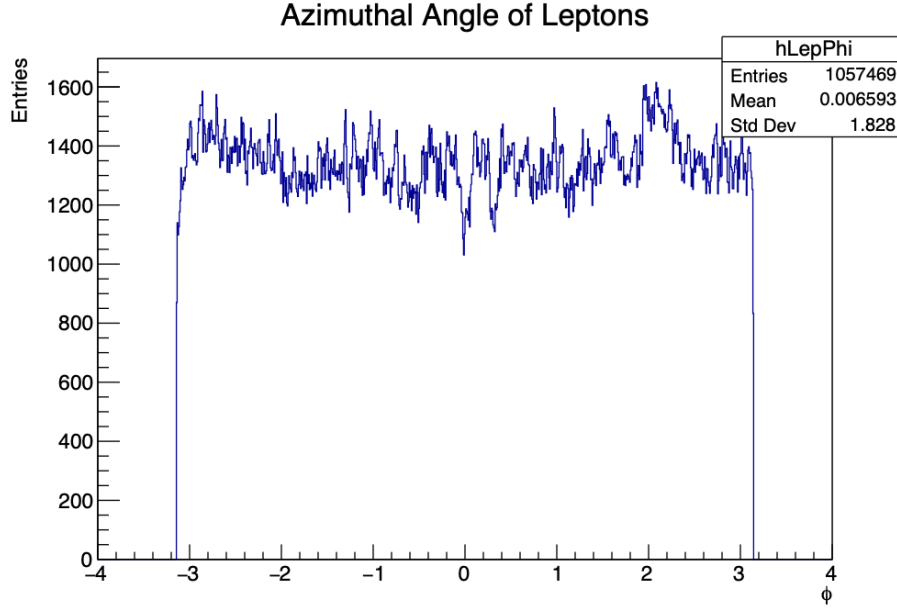


Figure 5: lep_phi graph

lep_phi is the Azimuthal angle of the lepton

Leptons are again distributed uniformly in azimuthal angle.

For graphing `lep_pt` and `lep_eta`, these are saved as vectors (conceptually `lists` like in `python`) because multiple leptons can be detected in one collision. Each one has its own momentum and pseudorapidity which are single components of the `lep_pt` and `lep_eta` vectors.

The histograms consolidate all of these measurements, so they can have more total entries/measurements than there were collisions.